

Chapter 18, Problem 12PP

Problem

An audio signal that is band-limited to 12.5 kHz is digitized into 8-bit samples. What is the maximum sampling interval that must be used to ensure complete recovery?

Step-by-step solution

Step 1 of 2

Given an audio signal is band limited to 12.5 kHz .

$$W = 12.5 \text{ kHz}$$

The minimum sampling frequency is the Nyquist frequency

$$f_s = 2W$$

$$f_s = 2 \times 12.5 \times 10^3$$

$$f_s = 25 \text{ kHz}$$

Step 2 of 2

The maximum sampling interval that must be used to ensure complete recovery is $T_s = \frac{1}{f_s}$

$$T_s = \frac{1}{25 \times 10^3}$$

$$T_s = 40 \times 10^{-6}$$

$$\therefore T_s = 40 \text{ } \mu\text{s}$$